UNIVERSITY OF PENNSYLVANIA
PROJECT 'IT'S PRONOUNCED EAGLES'

TEAM 90

VEHICLE SPECS

Vehicle: ELGSES

Length ---------- 383 cm

Diameter ---------- 15.7 cm

Mass ---------- 33.9 kg [wet]

Motor ---------- M2500

Target Alt. ---------- 3048 m

Projected Alt. ----------3199 m

Static Margin ---------- 2.21 cal

Thrust-to-Weight Ratio ------- 9.2 : 1

AVIONICS & COMMS

Flight Computer

SRAD custom flight computer, iterative upgrade on 2025 IREC design. Governs recovery, airbrakes, and payload.

Telemetry

Live telemetry & video downlink with custom ground-station UI. Yagi antenna for extended range.

Software

Flight computer software handles apogee detection, dual-deploy sequencing, and real-time AACS commands.

RECOVERY

Drogue at apogee · Main above avionics bay · Deployable payload deploys with drogue. All charges ground-tested.

PAYLOAD MISSIONS

Fixed Payload — POACH

POACH is a fixed 2U CubeSat-style payload housed in the nosecone. It uses onboard batteries, nichrome wire, insulation, and a sand thermal mass to first keep an egg cold before launch, then heat and cook it during flight after launch detection.

Deployable Payload — TALON

TALON is a 2U CubeSat-style deployable payload that separates from the rocket afore apogee. The mechanism uses a compact rack-and-pinion design integrated between the avionics bay sections.

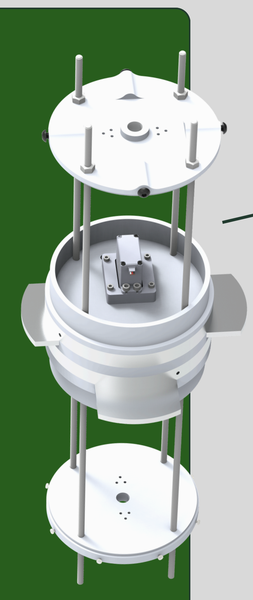


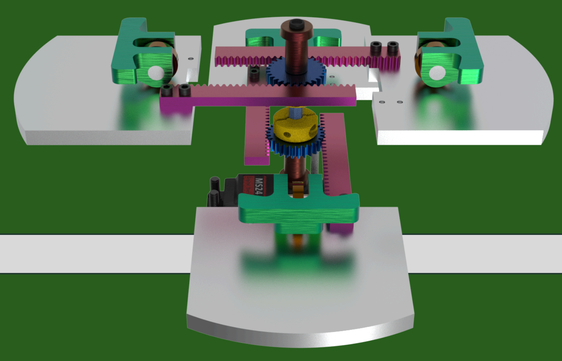


AIRBRAKES

The SRAD airbrake system actively controls apogee during coast. Four servo-driven aluminum leaves extend outward into the airflow, increasing drag and slowing the rocket before apogee.

The mechanism uses a compact rack-and-pinion design integrated between the avionics bay sections.



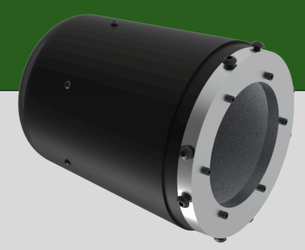


Colored for render differentiation

CUSTOM FINCAN

Four 3.2 mm G12 GFRE fins are root-bonded to the body tube with System 150 epoxy, using an aluminum jig for alignment and standard surface prep. External 25.4 mm fillets were hand-pulled after cure. Tip-to-tip layups (seven layers of 254 g/m² cloth + one layer of 102 g/m² cloth, pre-pregged with System 635 epoxy, alternating ±45°) add stiffness and improve load transfer.

The boat tail is 3D-printed in PETG-CF (chosen for heat resistance and in-house availability), with an internal machined aluminum tube carrying compressive loads — making the printed part a non-structural aeroskirt. It attaches to the fin can via screw fasteners.





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